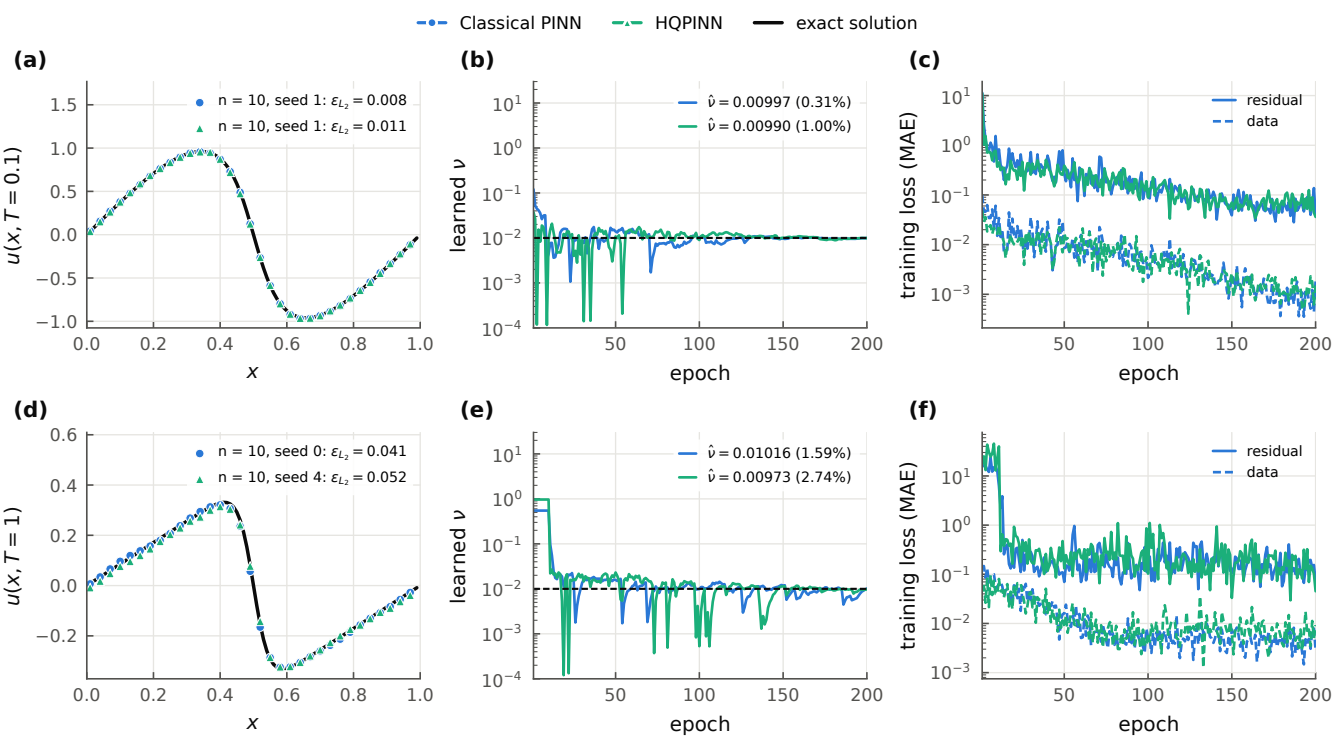

Classical PINN
HQPINN
exact solution
(a)
n = 10, seed 1: $\varepsilon_{L_2} = 0.008$
n = 10, seed 1: $\varepsilon_{L_2} = 0.011$
$u(x, T = 0.1)$
x
(b)
$\hat{\nu} = 0.00997$ (0.31%)
$\hat{\nu} = 0.00990$ (1.00%)
learned ν
epoch
(c)
residual
data
training loss (MAE)
epoch
(d)
n = 10, seed 0: $\varepsilon_{L_2} = 0.041$
n = 10, seed 4: $\varepsilon_{L_2} = 0.052$
$u(x, T = 1)$
x
(e)
$\hat{\nu} = 0.01016$ (1.59%)
$\hat{\nu} = 0.00973$ (2.74%)
learned ν
epoch
(f)
residual
data
training loss (MAE)
epoch